

SECTION 08 71 00 - DOOR HARDWARE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.
- B. All other related SPEC Sections are to be considered to complete the required product and services, including (but not limited to):
 - 1. 08 1113 – HOLLOW METAL DOORS AND FRAMES
 - 2. 08 1416 – FLUSH WOOD DOORS
 - 3. 08 4113 – ALUMINUM FRAMED ENTRANCES AND STOREFRONTS

1.2 SUMMARY

- A. This Section includes commercial door hardware for the following:
 - 1. Swinging doors.
- B. Door hardware includes, but is not necessarily limited to, the following:
 - 1. Mechanical door hardware.
 - 2. Electric access control hardware
 - 3. Cylinders specified for doors in other sections.
- C. Codes and References: Comply with the version year adopted by the Authority Having Jurisdiction.
 - 1. ANSI A117.1 - Accessible and Usable Buildings and Facilities.
 - 2. ICC/IBC - International Building Code.
 - 3. NFPA 70 - National Electrical Code.
 - 4. NFPA 80 - Fire Doors and Windows.
 - 5. NFPA 101 - Life Safety Code.
 - 6. NFPA 105 - Installation of Smoke Door Assemblies.
 - 7. State Building Codes, Local Amendments.
- D. Standards: All hardware specified herein shall comply with the following industry standards as applicable. Any undated reference to a standard shall be interpreted as referring to the latest edition of that standard:
 - 1. ANSI/BHMA Certified Product Standards - A156 Series.
 - 2. UL10C - Positive Pressure Fire Tests of Door Assemblies.

1.3 SUBMITTALS

- A. Product Data: Manufacturer's product data sheets including installation details, material descriptions, dimensions of individual components and profiles, operational descriptions and finishes.
- B. Door Hardware Schedule: Prepared by or under the supervision of supplier, detailing, fabrication and assembly of door hardware, as well as procedures and diagrams. Coordinate the final Door Hardware Schedule with doors, frames, and related work to ensure proper size, thickness, hand, function, and finish of door hardware.
 - 1. Format: Comply with scheduling sequence and vertical format in DHI's "Sequence and Format for the Hardware Schedule."
 - 2. Organization: Organize the Door Hardware Schedule into door hardware sets indicating complete designations of every item required for each door or opening. Organize door hardware sets in same order as in the Door Hardware Sets at the end of Part 3. Submittals that do not follow the same format and order as the Door Hardware Sets will be rejected and subject to resubmission.
 - 3. Content: Include the following information:
 - a. Type, style, function, size, label, hand, and finish of each door hardware item.
 - b. Manufacturer of each item.
 - c. Fastenings and other pertinent information.
 - d. Location of door hardware set, cross-referenced to Drawings, both on floor plans and in door and frame schedule.
 - e. Explanation of abbreviations, symbols, and codes contained in schedule.
 - f. Mounting locations for door hardware.
 - g. Door and frame sizes and materials.
 - h. Warranty information for each product.
 - 4. Submittal Sequence: Submit the final Door Hardware Schedule at earliest possible date, particularly where approval of the Door Hardware Schedule must precede fabrication of other work that is critical in the Project construction schedule. Include Product Data, Samples, Shop Drawings of other work affected by door hardware, and other information essential to the coordinated review of the Door Hardware Schedule.
- C. Keying Schedule: After a keying meeting with the owner has taken place prepare a separate keying schedule detailing final instructions. Submit the keying schedule in electronic format. Include keying system explanation, door numbers, key set symbols, hardware set numbers and special instructions. Owner must approve submitted keying schedule prior to the ordering of permanent cylinders/cores.
- D. Informational Submittals:
 - 1. Product Test Reports: Indicating compliance with cycle testing requirements, based on evaluation of comprehensive tests performed by manufacturer and witnessed by a qualified independent testing agency.

1.4 CLOSEOUT SUBMITTALS

- A. Operating and Maintenance Manuals: Provide manufacturers operating and maintenance manuals for each item comprising the complete door hardware installation in quantity as required in Division 01, Closeout Procedures.

- B. Project Record Documents: Provide record documentation of as-built door hardware sets in digital format (.pdf, .docx, .xlsx, .csv) and as required in Division 01, Project Record Documents.

1.5 QUALITY ASSURANCE

- A. Manufacturers Qualifications: Engage qualified manufacturers with a minimum 5 years of documented experience in producing hardware and equipment similar to that indicated for this Project and that have a proven record of successful in-service performance.
- B. Certified Products: Where specified, products must maintain a current listing in the Builders Hardware Manufacturers Association (BHMA) Certified Products Directory (CPD).
- C. Installer Qualifications: A minimum 3 years documented experience installing both standard and electrified door hardware similar in material, design, and extent to that indicated for this Project and whose work has resulted in construction with a record of successful in-service performance.
- D. Door Hardware Supplier Qualifications: Experienced commercial door hardware distributors with a minimum 5 years documented experience supplying both mechanical and electromechanical hardware installations comparable in material, design, and extent to that indicated for this Project. Supplier recognized as a factory direct distributor by the manufacturers of the primary materials with a warehousing facility in Project's vicinity. Supplier to have on staff a certified Architectural Hardware Consultant (AHC) available during the course of the Work to consult with Contractor, Architect, and Owner concerning both standard and electromechanical door hardware and keying.
- E. Source Limitations: Obtain each type and variety of door hardware specified in this section from a single source unless otherwise indicated.
 - 1. Electrified modifications or enhancements made to a source manufacturer's product line by a secondary or third party source will not be accepted.
- F. Each unit to bear third party permanent label indicating compliance with the referenced testing standards.
- G. Keying Conference: Conduct conference to comply with requirements in Division 01 Section "Project Meetings." Keying conference to incorporate the following criteria into the final keying schedule document:
 - 1. Function of building, purpose of each area and degree of security required.
 - 2. Plans for existing and future key system expansion.
 - 3. Requirements for key control storage and software.
 - 4. Installation of permanent keys, cylinder cores and software.
 - 5. Address and requirements for delivery of keys.
- H. Pre-Submittal Conference: Conduct coordination conference in compliance with requirements in Division 01 Section "Project Meetings" with attendance by representatives of Supplier(s), Installer(s), and Contractor(s) to review proper methods and the procedures for receiving, handling, and installing door hardware.
 - 1. Prior to installation of door hardware, conduct a project specific training meeting to instruct the installing contractors' personnel on the proper installation and adjustment of their respective products. Product training to be attended by installers of door hardware (including electromechanical hardware) for aluminum, hollow metal and wood doors.

- Training will include the use of installation manuals, hardware schedules, templates and physical product samples as required.
2. Inspect and discuss electrical roughing-in, power supply connections, and other preparatory work performed by other trades.
 3. Review sequence of operation narratives for each unique access controlled opening.
 4. Review and finalize construction schedule and verify availability of materials.
 5. Review the required inspecting, testing, commissioning, and demonstration procedures
- I. At completion of installation, provide written documentation that components were applied according to manufacturer's instructions and recommendations and according to approved schedule.

1.6 DELIVERY, STORAGE AND HANDLING

- A. Inventory door hardware on receipt and provide secure lock-up and shelving for door hardware delivered to Project site. Do not store electronic access control hardware, software or accessories at Project site without prior authorization.
- B. Tag each item or package separately with identification related to the final Door Hardware Schedule, and include basic installation instructions with each item or package.
- C. Deliver, as applicable, permanent keys, cylinders, cores, access control credentials, software and related accessories directly to Owner via registered mail or overnight package service. Instructions for delivery to the Owner shall be established at the "Keying Conference".

1.7 COORDINATION

- A. Templates: Obtain and distribute to the parties involved templates for doors, frames, and other work specified to be factory prepared for installing standard and electrified hardware. Check Shop Drawings of other work to confirm that adequate provisions are made for locating and installing hardware to comply with indicated requirements.
- B. Door and Frame Preparation: Doors and corresponding frames are to be prepared, reinforced and pre-wired (if applicable) to receive the installation of the specified electrified, monitoring, signaling and access control system hardware without additional in-field modifications.

1.8 WARRANTY

- A. General Warranty: Reference Division 01, General Requirements. Special warranties specified in this Article shall not deprive Owner of other rights Owner may have under other provisions of the Contract Documents and shall be in addition to, and run concurrent with, other warranties made by Contractor under requirements of the Contract Documents.
- B. Warranty Period: Written warranty, executed by manufacturer(s), agreeing to repair or replace components of standard and electrified door hardware that fails in materials or workmanship within specified warranty period after final acceptance by the Owner. Failures include, but are not limited to, the following:
 1. Structural failures including excessive deflection, cracking, or breakage.
 2. Faulty operation of the hardware.
 3. Deterioration of metals, metal finishes, and other materials beyond normal weathering.
 4. Electrical component defects and failures within the systems operation.

- C. Warranty Period: Unless otherwise indicated, warranty shall be one year from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PRODUCT SELECTION PROCEDURE (SPEC Section 01 60 00):

- A. Product: Where Specifications name a single manufacturer and product, provide the named product that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered.
- B. Manufacturer/Source: Where Specifications name a single manufacturer or source, provide a product by the named manufacturer or source that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered.
- C. Products:
 - 1. Restricted List: Where Specifications include a list of names of both manufacturers and products, provide one of the products listed that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered.
- D. Manufacturers:
 - 1. Restricted List: Where Specifications include a list of manufacturers' names, provide a product by one of the manufacturers listed that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered.
- E. Basis-of-Design Product: Where Specifications name a product, or refer to a product indicated on Drawings, and include a list of manufacturers, provide the specified or indicated product or a comparable product by one of the other named manufacturers. Drawings and Specifications indicate sizes, profiles, dimensions, and other characteristics that are based on the product named. Comply with requirements in "Comparable Products" Article for consideration of an unnamed product by one of the other named manufacturers.

2.2 BUTT HINGES

- A. Hinges: ANSI/BHMA A156.1 butt hinges with number of hinge knuckles and other options as specified in the Door Hardware Sets.
 - 1. Quantity: Provide the following hinge quantity:
 - a. Two Hinges: For doors with heights up to 60 inches.
 - b. Three Hinges: For doors with heights 61 to 90 inches.
 - c. Four Hinges: For doors with heights 91 to 120 inches.
 - d. For doors with heights more than 120 inches, provide 4 hinges, plus 1 hinge for every 30 inches of door height greater than 120 inches.
 - 2. Hinge Size: Provide the following, unless otherwise indicated, with hinge widths sized for door thickness and clearances required:
 - a. Widths up to 3'0": 4-1/2" standard or heavy weight as specified.

- b. Sizes from 3'1" to 4'0": 5" standard or heavy weight as specified.
- 3. Hinge Weight and Base Material: Unless otherwise indicated, provide the following:
 - a. Exterior Doors: Heavy weight, non-ferrous, ball bearing or oil impregnated bearing hinges unless Hardware Sets indicate standard weight.
 - b. Interior Doors: Standard weight, steel, ball bearing or oil impregnated bearing hinges unless Hardware Sets indicate heavy weight.
- 4. Hinge Options: Comply with the following:
 - a. Non-removable Pins: With the exception of electric through wire hinges, provide set screw in hinge barrel that, when tightened into a groove in hinge pin, prevents removal of pin while door is closed; for the all out-swinging lockable doors.
- 5. Manufacturers:
 - a. McKinney (MK) – MPB Series, 5-knuckle.

2.3 CONTINUOUS HINGES

- A. Continuous Geared Hinges: ANSI/BHMA A156.26 Grade 1-600 continuous geared hinge. with minimum 0.120-inch thick extruded 6063-T6 aluminum alloy hinge leaves and a minimum overall width of 4 inches. Hinges are non-handed, reversible and fabricated to template screw locations. Factory trim hinges to suit door height and prepare for electrical cut-outs.
 - 1. Manufacturers:
 - a. Pemko (PE).

2.4 DOOR OPERATING TRIM

- A. Flush Bolts and Surface Bolts: Provide products conforming to ANSI/BHMA A156.3 and A156.16, Grade 1.
 - 1. Flush bolts to be furnished with top rod of sufficient length to allow bolt retraction device location approximately six feet from the floor.
 - 2. Furnish dust proof strikes for bottom bolts.
 - 3. Surface bolts to be minimum 8" in length and U.L. listed for labeled fire doors and U.L. listed for windstorm components where applicable.
 - 4. Provide related accessories (mounting brackets, strikes, coordinators, etc.) as required for appropriate installation and operation.
 - 5. Manufacturers:
 - a. Rockwood (RO).
- B. Door Push Plates and Pulls: ANSI/BHMA A156.6 door pushes and pull units of type and design specified in the Hardware Sets. Coordinate and provide proper width and height as required where conflicting hardware dictates.
 - 1. Push/Pull Plates: Minimum .050 inch thick, size as indicated in hardware sets, with beveled edges, secured with exposed screws unless otherwise indicated.

2. Door Pull and Push Bar Design: Size, shape, and material as indicated in the hardware sets. Minimum clearance of 2 1/2-inches from face of door unless otherwise indicated.
 3. Offset Pull Design: Size, shape, and material as indicated in the hardware sets. Minimum clearance of 2 1/2-inches from face of door and offset of 90 degrees unless otherwise indicated.
 4. Pulls, where applicable, shall be provided with a 10" clearance from the finished floor on the push side to accommodate wheelchair accessibility.
 5. Fasteners: Provide manufacturer's designated fastener type as indicated in Hardware Sets. When through-bolt fasteners are in the same location as a push plate, countersink the fasteners flush with the door face allowing the push plate to sit flat against the door.
 6. Manufacturers:
 - a. Rockwood (RO).
- C. Locking Pull System: Post-mount style door pulls with integrated deadbolt locking system in type and design as specified in the Hardware Sets. Pulls available in multiple head, floor, or combination locking options, with outside keyed rim cylinder operation and inside turn piece activation. Mounting applications for aluminum, glass, steel and wood doors, with customized sizing and configuration options. Locking pulls shall be provided with a 10" clearance from the finished floor on the cylinder side to accommodate wheelchair accessibility.
1. Manufacturers:
 - a. Rockwood (RO) - LP Series.

2.5 CYLINDERS AND KEYING

- A. General: Cylinder manufacturer to have minimum (10) years experience designing secured master key systems and have on record a published security keying system policy.
- B. Cylinder Types: Original manufacturer cylinders able to supply the following cylinder formats and types:
 1. Threaded mortise cylinders with rings and cams to suit hardware application.
 2. Rim cylinders with back plate, flat-type vertical or horizontal tailpiece, and raised trim ring.
 3. Bored or cylindrical lock cylinders with tailpieces as required to suit locks.
 4. Tubular deadlocks and other auxiliary locks.
 5. Mortise and rim cylinder collars to be solid and recessed to allow the cylinder face to be flush and be free spinning with matching finishes.
 6. Keyway: **TBD**
- C. Patented Cylinders: ANSI/BHMA A156.5, Grade 1 Certified Products Directory (CPD) listed cylinders employing a utility patented and restricted keyway requiring the use of a patented key. Cylinders are to be protected from unauthorized manufacture and distribution by manufacturer's United States patents. Cylinders are to be factory keyed with owner having the ability for on-site original key cutting.
 1. Patented key systems shall not be established with products that have an expired patent. Expired systems shall only be specified and supplied to support existing systems.
 2. Manufacturers:
 - a. **MEDECO X4 (SFIC)**
 - 1) **VERIFY PROPOSED KEY SYSTEM WITH END USER**

- D. Keying System: Each type of lock and cylinders to be factory keyed.
 - 1. Supplier shall conduct a "Keying Conference" to define and document keying system instructions and requirements.
 - 2. Furnish factory cut, nickel-silver large bow permanently inscribed with a visual key control number as directed by Owner.
 - 3. New System: Key locks to a new key system as directed by the Owner.
- E. Key Quantity: Provide the following minimum number of keys:
 - 1. Change Keys per Cylinder: Two (2)
 - 2. Master Keys (per Master Key Level/Group): Five (5).
 - 3. Construction Keys (where required): Ten (10).
- F. Construction Keying: Provide construction use interchangeable cores
- G. Key Registration List (Bitting List):
 - 1. Provide keying transcript list to Owner's representative in the proper format for importing into key control software.
 - 2. Provide transcript list in writing or electronic file as directed by the Owner.

2.6 KEY CONTROL

- A. Key Control Cabinet: Provide a key control system including envelopes, labels, and tags with self-locking key clips, receipt forms, 3-way visible card index, temporary markers, permanent markers, and standard metal cabinet. Key control cabinet shall have expansion capacity of 150% of the number of locks required for the project.
 - 1. Manufacturers:
 - a. Lund Equipment (LU).
 - b. MMF Industries (MM).
 - c. Telkee (TK).

2.7 CYLINDRICAL LOCKS AND LATCHING DEVICES

- A. Cylindrical Locksets, Grade 1 (Heavy Duty): Provide ANSI/BHMA A156.2, Series 4000, Operational Grade 1 Certified Products Directory (CPD) listed cylindrical locksets. Listed manufacturers shall meet all functions and features as specified herein.
 - 1. Manufacturers:
 - a. Sargent Manufacturing (SA) – 10X Series.

2.8 LOCK AND LATCH STRIKES

- A. Strikes: Provide manufacturer's standard strike with strike box for each latch or lock bolt, with curved lip extended to protect frame, finished to match door hardware set, unless otherwise indicated, and as follows:

1. Flat-Lip Strikes: For locks with three-piece antifriction latch bolts, as recommended by manufacturer.
2. Extra-Long-Lip Strikes: For locks used on frames with applied wood casing trim.
3. Aluminum-Frame Strike Box: Provide manufacturer's special strike box fabricated for aluminum framing.
4. Double-lipped strikes: For locks at double acting doors. Furnish with retractable stop for rescue hardware applications.

B. Standards: Comply with the following:

1. Strikes for Mortise Locks and Latches: BHMA A156.13.
2. Strikes for Bored Locks and Latches: BHMA A156.2.
3. Strikes for Auxiliary Deadlocks: BHMA A156.36.
4. Dustproof Strikes: BHMA A156.16.

2.9 CONVENTIONAL EXIT DEVICES

A. General Requirements: All exit devices specified herein shall meet or exceed the following criteria:

1. Exit devices shall have a five-year warranty.
2. At doors not requiring a fire rating, provide devices complying with NFPA 101 and listed and labeled for "Panic Hardware" according to UL305. Provide proper fasteners as required by manufacturer including sex nuts and bolts at openings specified in the Hardware Sets.
3. Where exit devices are required on fire rated doors, provide devices complying with NFPA 80 and with UL labeling indicating "Fire Exit Hardware". Provide devices with the proper fasteners for installation as tested and listed by UL. Consult manufacturer's catalog and template book for specific requirements.
4. Except on fire rated doors, provide exit devices with hex key dogging device to hold the push bar and latch in a retracted position. Provide optional keyed cylinder dogging on devices where specified in Hardware Sets.
5. Devices must fit flat against the door face with no gap that permits unauthorized dogging of the push bar. The addition of filler strips is required in any case where the door light extends behind the device as in a full glass configuration.
6. Lever Operating Trim: Where exit devices require lever trim, furnish manufacturer's heavy duty escutcheon trim with threaded studs for thru-bolts.
 - a. Lock Trim Design: As indicated in Hardware Sets, provide finishes and designs to match that of the specified locksets.
 - b. Where function of exit device requires a cylinder, provide a cylinder (Rim or Mortise) as specified in Hardware Sets.
7. Vertical Rod Exit Devices: Where surface or concealed vertical rod exit devices are used at interior openings, provide as less bottom rod (LBR) unless otherwise indicated. Provide dust proof strikes where thermal pins are required to project into the floor.
8. Narrow Stile Applications: At doors constructed with narrow stiles, or as specified in Hardware Sets, provide devices designed for maximum 2" wide stiles.
9. Dummy Push Bar: Nonfunctioning push bar matching functional push bar.
10. Rail Sizing: Provide exit device rails factory sized for proper door width application.
11. ~~Through Bolt Installation: For exit devices and trim as indicated in Door Hardware Sets.~~

a. **Through bolting is not acceptable**

- B. Conventional Push Rail Exit Devices (Heavy Duty): ANSI/BHMA A156.3, Grade 1 Certified Products Directory (CPD) listed exit devices. Listed manufacturers shall meet all functions and features as specified herein.
1. Provide exit devices with functions and features as follows:
 - a. Where required by code, provide knurling or abrasive coating on all levers leading to hazardous areas.
 - b. Meets UL and CUL Standard 10C Positive Pressure, Fire Test of Door Assemblies with levers that meet A117.1 Accessibility Code.
 - c. No catch points: addition of applied deflectors or other added components are not allowed.
 - d. No visible plastic.
 - e. Heavy duty end caps with flush and overlapping options made of stainless steel, brass, or bronze with architectural finishes.
 - f. Constructed of all stainless steel.
 - g. Stainless steel pullman type latch with deadlock feature.
 - h. Narrow or wide style exterior trim as specified in the hardware sets.
 - i. Center case adjustability on concealed vertical rod exit devices; single operation with hex key individually adjusts top or bottom latches. No retainer screws or clips required to maintain adjustment.
 - j. Ten-year limited warranty for mechanical features.
 2. Manufacturers:
 - a. Sargent Manufacturing (SA) - PE80 Series.

2.10 DOOR CLOSERS

- A. All door closers specified herein shall meet or exceed the following criteria:
1. General: Door closers to be from one manufacturer, matching in design and style, with the same type door preparations and templates regardless of application or spring size. Closers to be non-handed with full sized covers.
 2. Standards: Closers to comply with UL-10C for Positive Pressure Fire Test and be U.L. listed for use of fire rated doors.
 3. Size of Units: Comply with manufacturer's written recommendations for sizing of door closers depending on size of door, exposure to weather, and anticipated frequency of use. Where closers are indicated for doors required to be accessible to the Americans with Disabilities Act, provide units complying with ANSI ICC/A117.1.
 4. Closer Arms: Provide heavy duty, forged steel closer arms unless otherwise indicated in Hardware Sets.
 5. Closers shall not be installed on exterior or corridor side of doors; where possible install closers on door for optimum aesthetics.
 6. Closer Accessories: Provide door closer accessories including custom templates, special mounting brackets, spacers and drop plates as required for proper installation. ~~Provide through bolt and security type fasteners as specified in the hardware sets.~~
 - a. **Through bolting is not acceptable**
- B. Door Closers, Overhead Concealed Mounted (Commercial Duty): ANSI/BHMA 156.4, Grade 1 Certified Products Directory (CPD) listed surface mounted, institutional grade door closers with complete spring power adjustment, sizes 1 thru 6; and fully operational adjustable according to door size, frequency of use, and opening force. Closers to be rack and pinion type, one piece

cast iron or aluminum alloy body construction, with adjustable backcheck, closing sweep, and latch speed control valves. Provide non-handed units standard.

1. Manufacturers:
 - a. Sargent Manufacturing (SA)
 - 1) 268 & 269 Overhead Concealed Closers (as required per hardware sets)
 - 2) 351 Series (as required per hardware sets)
 - 3) 1131 Series (as required per hardware sets)

2.11 ARCHITECTURAL TRIM

A. Door Protective Trim

1. General: Door protective trim units to be of type and design as specified below or in the Hardware Sets.
2. Size: Fabricate protection plates (kick, armor, or mop) not more than 2" less than door width (LDW) on stop side of single doors and 1" LDW on stop side of pairs of doors, and not more than 1" less than door width on pull side. Coordinate and provide proper width and height as required where conflicting hardware dictates. Height to be as specified in the Hardware Sets.
3. Where plates are applied to fire rated doors with the top of the plate more than 16" above the bottom of the door, provide plates complying with NFPA 80. Consult manufacturer's catalog and template book for specific requirements for size and applications.
4. Protection Plates: ANSI/BHMA A156.6 protection plates (kick, armor, or mop), fabricated from the following:
 - a. Stainless Steel: 300 grade, 050-inch thick.
5. Options and fasteners: Provide manufacturer's designated fastener type as specified in the Hardware Sets. Provide countersunk screw holes.
6. Manufacturers:
 - a. Rockwood (RO).

2.12 DOOR STOPS AND HOLDERS

- A. General: Door stops and holders to be of type and design as specified below or in the Hardware Sets.
- B. Door Stops and Bumpers: ANSI/BHMA A156.16, Grade 1 door stops and wall bumpers. Provide wall bumpers, either convex or concave types with anchorage as indicated, unless floor or other types of door stops are specified in Hardware Sets. Do not mount floor stops where they will impede traffic. Where floor or wall bumpers are not appropriate, provide overhead type stops and holders.
 1. Manufacturers:
 - a. Rockwood (RO).
- C. Overhead Door Stops and Holders: ANSI/BHMA A156.8, Grade 1 Certified Products Directory (CPD) listed overhead stops and holders to be surface or concealed types as indicated in Hardware Sets. Track, slide, arm and jamb bracket to be constructed of extruded bronze and

shock absorber spring of heavy tempered steel. Provide non-handed design with mounting brackets as required for proper operation and function.

1. Manufacturers:
 - a. Sargent Manufacturing (SA).

2.13 ARCHITECTURAL SEALS

- A. General: Thresholds, weatherstripping, and gasket seals to be of type and design as specified below or in the Hardware Sets. Provide continuous weatherstrip gasketing on exterior doors and provide smoke, light, or sound gasketing on interior doors where indicated. At exterior applications provide non-corrosive fasteners and elsewhere where indicated.
- B. Smoke Labeled Gasketing: Assemblies complying with NFPA 105 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for smoke control ratings indicated, based on testing according to UL 1784.
 1. Provide smoke labeled perimeter gasketing at all smoke labeled openings.
- C. Fire Labeled Gasketing: Assemblies complying with NFPA 80 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire ratings indicated, based on testing according to UL-10C.
 1. Provide intumescent seals as indicated to meet UL10C Standard for Positive Pressure Fire Tests of Door Assemblies, and NFPA 252, Standard Methods of Fire Tests of Door Assemblies.
- D. Sound-Rated Gasketing: Assemblies that are listed and labeled by a testing and inspecting agency, for sound ratings indicated.
- E. Replaceable Seal Strips: Provide only those units where resilient or flexible seal strips are easily replaceable and readily available from stocks maintained by manufacturer.
- F. Manufacturers:
 1. Pemko (PE).

2.14 FABRICATION

- A. Fasteners: Provide door hardware manufactured to comply with published templates generally prepared for machine, wood, and sheet metal screws. Provide screws according to manufacturers recognized installation standards for application intended.

2.15 FINISHES

- A. Standard: Designations used in the Hardware Sets and elsewhere indicate hardware finishes complying with ANSI/BHMA A156.18, including coordination with traditional U.S. finishes indicated by certain manufacturers for their products.

- B. Provide quality of finish, including thickness of plating or coating (if any), composition, hardness, and other qualities complying with manufacturer's standards, but in no case less than specified by referenced standards for the applicable units of hardware
- C. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine scheduled openings, with Installer present, for compliance with requirements for installation tolerances, labeled fire door assembly construction, wall and floor construction, and other conditions affecting performance.
- B. Notify architect of any discrepancies or conflicts between the door schedule, door types, drawings and scheduled hardware. Proceed only after such discrepancies or conflicts have been resolved in writing.

3.2 PREPARATION

- A. Hollow Metal Doors and Frames: Comply with ANSI/DHI A115 series.
- B. Wood Doors: Comply with ANSI/DHI A115-W series.

3.3 INSTALLATION

- A. Install each item of mechanical and electromechanical hardware and access control equipment to comply with manufacturer's written instructions and according to specifications.
 - 1. Installers are to be trained and certified by the manufacturer on the proper installation and adjustment of fire, life safety, and security products including: hanging devices; locking devices; closing devices; and seals.
- B. Mounting Heights: Mount door hardware units at heights indicated in following applicable publications, unless specifically indicated or required to comply with governing regulations:
 - 1. Standard Steel Doors and Frames: DHI's "Recommended Locations for Architectural Hardware for Standard Steel Doors and Frames."
 - 2. DHI TDH-007-20: Installation Guide for Doors and Hardware.
 - 3. Where indicated to comply with accessibility requirements, comply with ANSI A117.1 "Accessibility Guidelines for Buildings and Facilities."
 - 4. Provide blocking in drywall partitions where wall stops or other wall mounted hardware is located.
- C. Retrofitting: Install door hardware to comply with manufacturer's published templates and written instructions. Where cutting and fitting are required to install door hardware onto or into surfaces that are later to be painted or finished in another way, coordinate removal, storage, and reinstallation of surface protective trim units with finishing work specified in Division 9 Sections. Do not install surface-mounted items until finishes have been completed on substrates involved.

- D. Push Plates and Door Pulls: When through-bolt fasteners are in the same location as a push plate, countersink the fasteners flush with the door face allowing the push plate to sit flat against the door.
- E. Thresholds: Set thresholds for exterior and acoustical doors in full bed of sealant complying with requirements specified in Division 7 Section "Joint Sealants."
- F. Storage: Provide a secure lock up for hardware delivered to the project but not yet installed. Control the handling and installation of hardware items so that the completion of the work will not be delayed by hardware losses before and after installation.

3.4 FIELD QUALITY CONTROL

- A. Field Inspection (Punch Report): Reference Division 01 Sections "Closeout Procedures". Produce project punch report for each installed door opening indicating compliance with approved submittals and verification hardware is properly installed, operating and adjusted. Include list of items to be completed and corrected, indicating the reasons or deficiencies causing the Work to be incomplete or rejected.
 - 1. Organization of List: Include separate Door Opening and Deficiencies and Corrective Action Lists organized by Mark, Opening Remarks and Comments, and related Opening Images and Video Recordings.

3.5 ADJUSTING

- A. Initial Adjustment: Adjust and check each operating item of door hardware and each door to ensure proper operation or function of every unit. Replace units that cannot be adjusted to operate as intended. Adjust door control devices to compensate for final operation of heating and ventilating equipment and to comply with referenced accessibility requirements.

3.6 CLEANING AND PROTECTION

- A. Protect all hardware stored on construction site in a covered and dry place. Protect exposed hardware installed on doors during the construction phase. Install any and all hardware at the latest possible time frame.
- B. Clean adjacent surfaces soiled by door hardware installation.
- C. Clean operating items as necessary to restore proper finish. Provide final protection and maintain conditions that ensure door hardware is without damage or deterioration at time of owner occupancy.

3.7 DEMONSTRATION

- A. Instruct Owner's maintenance personnel to adjust, operate, and maintain mechanical and electromechanical door hardware.

3.8 DOOR HARDWARE SETS

- A. The hardware sets represent the design intent and direction of the owner and architect. They are a guideline only and should not be considered a detailed hardware schedule. Discrepancies, conflicting hardware and missing items should be brought to the attention of the architect with corrections made prior to the bidding process. Omitted items not included in a hardware set should be scheduled with the appropriate additional hardware required for proper application and functionality.
1. Quantities listed are for each pair of doors, or for each single door.
 2. The supplier is responsible for handing and sizing all products.
 3. Where multiple options for a piece of hardware are given in a single line item, the supplier shall provide the appropriate application for the opening.

B. Manufacturer's Abbreviations:

1. AD Adams Rite
2. BE BEST (Construction cores & keys)
3. HS HES
4. MC McKinney
5. MED1 Medeco
6. PE Pemko
7. RO Rockwood
8. SA Sargent
9. VA01 Various MFG (Key cabinet)

C. Hardware Options/Codes:

Code	Description
1 3/4" DOORS	1-3/4" Thick Doors
13-0512	ADAMS RITE CAM (AR1850) STD CYLS
2 3/4" BS	2 3/4" BACKSET
2" Doors	2" Thick Doors
2004M	Electrolynx Adaptor
31	FOR DOORS UP TO 2 1/4"
70	Small Format Construction Plastic Core
7P	7 Pin
CTC	CENTER TO CENTER MOUNTING MEASUREMENT
Door Height	Door Height in Inches (Specify)
LC	LESS CYLINDER-SINGLE (RIM/MORTISE)
MSES25SS	SS/MS&ES25 (for 36")
MSES25SS	SS/MS&ES25 (for 48")
MSES25SS	SS/MS&ES25 (for 72")
OVERALL LENGTH	OVERALL MEASUREMENT
SRI	RUST INHIBITOR PAINT-ARMS & BODY
Straight Bolt	Straight Bolt
TB	MS and Mortise Nuts for Closers
TYPE 12XHD FASTENING	DECORATIVE HD TB MOUNTING (3/8-16)
V21	Ind Both Sds Lckd/Unlckd x Vcnt/Occ, G/R

D. Hardware Finish List

Code	Description
10BE	Dark Oxidized Satin Bronze – Equivalent
313	Dark Bronze Duranodic
613E	Dark Oxidized Satin Bronze – Equivalent
630	Satin Stainless Steel (placeholder for 5000 electric strike)
BLK	Black
EB	Sprayed Finish, To Match 10B

All exposed hardware, fasteners, etc. shall be 10BE as the BOD finish color. If 10BE is not an option, submit an alternative finish for the design team to review and approve.

E. Hardware Sets:

Set #101

- 1 Pair Doors #101 Vesitbule 103 to/from Exteri- 105° RHRA
or

Opening Description: 2 - 3' 0" x 8' 0" Type AL-NA-D8 Type AL-NA x N/A

2	Continuous Hinge	K10BEFM95HD1		PE
1	Deadlock	MS1850S Door Height Straight Bolt	313	AD
	Properties: Door Height=96			
1	Thumbturn Cylinder	4066-01	313	AD
1	Mortise Cylinder	70 7P 43 13-0512	10BE	SA
	Note: - SFIC Cylinder housing with disposable core			
1	Construction Core	7190224		BE
	Note: - SFIC Construction core			
2	Pull	RM3301 35" 2" Doors CTC OVERALL LENGTH	10BE	RO
	Note: - Push bar dimensions shall be coordinated with Aluminum Door provider & approved by the deisgn team during submittal phase			
	Properties: CTC = =33, Overall Length=35			
2	Pull	RM3310 30" 2" Doors CTC OVERALL LENGTH	10BE	RO
	Note: - Verify door thickness			
1	Closer	SRI 268 OB	RH 10BE	SA
1	Closer	SRI 268 OB	LH 10BE	SA
1	SFIC	33700006 N 26 DAKM		MED1
	Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
	- End user needs to confirm master key system & keyway prior procurement			
1	Threshold	2008-10BE-PK 72" MSES25SS		PE
	Note: 1. Integrated Seals (door bottoms, astragals, etc) by Aluminum Door/Frames Provider			
	2. Closer drop plates and spacers are not specified in the hardware sets. Aluminum provider shall coordinate if they are required to complete their division.			
	3. Aluminum Opening Hardware finish color shall be determined by the Architect at a later time			

Set #102

1 Single Door #102 Stair #01 Vestibule 103 to/from Stair 105° LHR
#01 102

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-60-D5 x HMF Type HM-60-F1 x
60MIN

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Exit Device	LC PE8806 F WEMD	LHR 10BE	SA
1 Rim Cylinder	70 7P 34	10BE	SA
Note: - SFIC Cylinder housing with disposable core			
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Electric Strike	9600-LBM 2004M	613E	HS
1 Closer	351 CPS TB	10BE	SA
1 Wiring Harness	QC-C1500		MC
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
1 Power Supply	LSP-12-24		SA
1 Smoke Seal	S88 BL 21'		PE

Set #103

1 Pair Doors #103

Vestibule 103 to/from Bank- 105° RHRA
ing Personal 106

Opening Description: 2 - 3' 0" x 8' 0" Type AL-NA-D8 Type AL-NA x N/A

2	Continuous Hinge	K10BEFM95HD1		PE
1	Deadlock	MS1850S Door Height Straight Bolt	313	AD
	Properties: Door Height=96			
1	Thumbturn Cylinder	4066-01	313	AD
1	Mortise Cylinder	70 7P 43 13-0512	10BE	SA
	Note: - SFIC Cylinder housing with disposable core			
1	Construction Core	7190224		BE
	Note: - SFIC Construction core			
2	Pull	RM3301 35" 2" Doors CTC OVERALL LENGTH	10BE	RO
	Note: - Push bar dimensions shall be coordinated with Aluminum Door provider & approved by the design team during submittal phase			
	Properties: CTC = 33, Overall Length=35			
2	Pull	RM3310 30" 2" Doors CTC OVERALL LENGTH	10BE	RO
	Note: - Verify door thickness			
1	Closer	SRI 268 OB	RH 10BE	SA
1	Closer	SRI 268 OB	LH 10BE	SA
2	Dome Stop	441	10BE	RO
1	SFIC	33700006 N 26 DAKM		MED1
	Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
	- End user needs to confirm master key system & keyway prior procurement			
	Note: 1. Integrated Seals (door bottoms, astragals, etc) by Aluminum Door/Frames Provider			
	2.			
	3.			
	4. Closer drop plates and spacers are not specified in the hardware sets. Aluminum provider shall coordinate if they are required to complete their division.			

Set #104

1 Single Door #104

Banking Personal 106 to/from 105° RH
Office 104

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D3 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG05 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #105

1 Single Door #105

Banking Personal 106 to/from 105° RH
Office 105

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D3 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG05 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #106

1 Single Door #106

Banking Personal 106 to/from 105° LHR
Exterior

Opening Description: 3' 0" x 8' 0" Type AL-NA-D9 Type AL-NA x N/A

1 Continuous Hinge	K10BEFM95HD1			PE
1 Exit Device	31 PE8804 F LESS TRIM	LHR	10BE	SA
Note: - Verify door thickness for code "31"				
Properties: Notes=TBD				
1 Rim Cylinder	70 7P 34		10BE	SA
Note: - SFIC Cylinder housing with disposable core				
1 Construction Core	7190224			BE
Note: - SFIC Construction core				
1 Electric Strike	9600-LBM 2004M		613E	HS
1 Pull	RM3310 30" 2" Doors CTC OVERALL LENGTH		10BE	RO
Note: - Verify door thickness				
1 Closer	SRI 268 OB	RH	10BE	SA
1 Dome Stop	441		10BE	RO
1 Wiring Harness	QC-C1500			MC
1 SFIC	33700006 N 26 DAKM			MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections				
- End user needs to confirm master key system & keyway prior procurement				
1 Power Supply	LSP-12-24			SA
1 Threshold	2008-10BE-PK 36" MSES25SS			PE

Set #107

1 Single Door #107

Banking Personal 106 to/from 105° LH
Office 107

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D3 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG05 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #108

1 Single Door #108

Banking Personal 106 to/from 105° RH
Office 108

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D3 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG05 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #109

1 Single Door #109

Banking Personal 106 to/from 105° RH
Office 109

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D3 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG05 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #111

1 Single Door #111

Teller 112 to/from Cash RM 105° LH
111

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D4 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG37 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Electric Strike	5000C-LBM 2004M	630	HS
1 Closer	1131 HU	EB	SA
1 Wall Stop	409	10BE	RO
1 Wiring Harness	QC-C1500		MC
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
1 Power Supply	LSP-12-24		SA
3 Door Silencer	608-RKW	BLK	RO

Set #112

1 Single Door #112

Banking Personal 106 to/from 105° RH
Teller 112

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D2 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG04 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Electric Strike	5000C-LBM 2004M	630	HS
Note: - Product finish shall be 613 per product catalog			
1 Closer	1131 RU	EB	SA
1 Wall Stop	409	10BE	RO
1 Wiring Harness	QC-C1500		MC
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
1 Power Supply	LSP-12-24		SA
3 Door Silencer	608-RKW	BLK	RO

Set #113

1 Single Door #113

Corridor 114 to/from Women 105° RH
113

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D1 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Deadbolt	468 V21	10BE	SA
1 Keypad Lock	70 KP10XG77 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Closer	1131 RU	EB	SA
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #115

1 Single Door #115

Corridor 114 to/from Men
115

105° LH

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D1 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Deadbolt	468 V21	10BE	SA
1 Keypad Lock	70 KP10XG77 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Closer	1131 RU	EB	SA
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #116

1 Single Door #116

Elevator Equip 116 to/from 105° LHR
Exterior

Opening Description: 4' 0" x 7' 10" x 1 3/4" x HMD Type HM-NA-D7 x HMF Type HM-NA-F1 x
N/A

1 Continuous Hinge	10BEFM95HD1		PE
1 Lockset	70 10XG04 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Closer	351 CPS	10BE	SA
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
1 Weatherstrip	18061-10BE-NB 48"		PE
1 Raindrip	346-10BE 52"		PE
1 Weatherstrip	290-10BE-PK 1 x 48" 2 x 96"		PE
1 Threshold	2008-10BE-PK 48" MSSES25SS		PE
3 Door Silencer	608-RKW	BLK	RO

Set #117

1 Single Door #117

Corridor 114 to/from Electrical 117 105° LHR

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-90-D1 x HMF Type HM-90-F1 x 90Min

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG04 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Electric Strike	5000C-LBM 2004M	630	HS
1 Closer	1131 RU	EB	SA
1 Wall Stop	409	10BE	RO
1 Wiring Harness	QC-C1500		MC
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
1 Power Supply	LSP-12-24		SA
1 Smoke Seal	S88 BL 21'		PE

Set #118

1 Single Door #118

Banking Personal 106 to/from 105° RH
Storage 118

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D1 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG37 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #119

1 Single Door #119

Stair #2 119 to/from Exterior 105° LHR

Opening Description: 3' 0" x 7' 10" x 1 3/4" x HMD Type HM-NA-D7 x HMF Type HM-NA-F1 x
N/A

1 Continuous Hinge	10BEFM95HD1			PE
1 Exit Device	LC PE8806 F WEMD	LHR	10BE	SA
1 Rim Cylinder	70 7P 34		10BE	SA
Note: - SFIC Cylinder housing with disposable core				
1 Construction Core	7190224			BE
Note: - SFIC Construction core				
1 Electric Strike	9600-LBM 2004M		613E	HS
1 Closer	351 CPS		10BE	SA
1 Wiring Harness	QC-C1500			MC
1 SFIC	33700006 N 26 DAKM			MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections				
- End user needs to confirm master key system & keyway prior procurement				
1 Power Supply	LSP-12-24			SA
1 Weatherstrip	290-10BE-PK 1 x 36" 2 x 96"			PE
1 Raindrip	346-10BE 40"			PE
1 Weatherstrip	18061-10BE-NB 36"			PE
1 Threshold	2008-10BE-PK 36" MSES25SS			PE

Set #201

1 Single Door #201 Stair #01 Stair #01 201 to/from Vestibule 202 105° LHR

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-60-D5 x HMF Type HM-60-F1 x 60MIN

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Passage Set	10XU15 LMD 2 3/4" BS	10BE	SA
1 Closer	351 CPS TB	10BE	SA
1 Wall Stop	409	10BE	RO
1 Smoke Seal	S88 BL 21'		PE

Set #203

1 Single Door #203

Open Space 206 to/from Of- 105° LH
fice 203

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D3 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG05 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #205

1 Pair Doors #205

Vestibule 204 to/from Board 105° RHRA
RM 205

Opening Description: 2 - 3' 0" x 8' 0" x 1 3/4" x WD Type PL-NA-D6 x HMF Type HM-NA-F2 x
N/A

8 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
2 Straight Pull	RM3101 30" 1 3/4" DOORS CTC	10BE	RO
	OVERALL LENGTH TYPE 12XHD FASTENING		
Properties: CTC = 24, Overall Length=30			
2 Cast Plate	RM5596B 3 1/2 x 15 1 3/4" DOORS	10BE	RO
1 Closer	268 HB	RH 10BE	SA
1 Closer	268 HB	LH 10BE	SA
2 Door Silencer	608-RKW	BLK	RO

Set #207

1 Single Door #207

Open Space 206 to/from Of- 105° RH
fice 207

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D3 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG05 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #208

1 Single Door #208

Open Space 206 to/from Of- 105° RH
fice 208

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D3 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG05 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #210

1 Single Door #210

Corridor 213 to/from Janitor 105° RHR
210

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D1 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG04 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Overhead Stop	598S	10BE	SA
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #211

1 Single Door #211

Corridor 213 to/from Break 105° LH
RM 211

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D2 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Passage Set	10XU15 LMD 2 3/4" BS	10BE	SA
1 Closer	1131 HU	EB	SA
1 Wall Stop	409	10BE	RO
3 Door Silencer	608-RKW	BLK	RO

Set #212

1 Single Door #212

Corridor 213 to/from Stair
#02 212

105° LHR

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-60-D5 x HMF Type HM-60-F1 x
60MIN

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Passage Set	10XU15 LMD 2 3/4" BS	10BE	SA
1 Closer	351 CPS TB	10BE	SA
1 Wall Stop	409	10BE	RO
1 Smoke Seal	S88 BL 21'		PE

Set #214

1 Single Door #214

Corridor 213 to/from Men 105° LH
214

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D1 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Deadbolt	468 V21	10BE	SA
1 Passage Set	10XU15 LMD 2 3/4" BS	10BE	SA
1 Closer	1131 RU	EB	SA
1 Wall Stop	409	10BE	RO
3 Door Silencer	608-RKW	BLK	RO

Set #215

1 Single Door #215

Corridor 213 to/from Women 105° RH
215

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D1 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Deadbolt	468 V21	10BE	SA
1 Passage Set	10XU15 LMD 2 3/4" BS	10BE	SA
1 Closer	1131 RU	EB	SA
1 Wall Stop	409	10BE	RO
3 Door Silencer	608-RKW	BLK	RO

Set #216

1 Single Door #216

Open Space 206 to/from Stor- 105° RH
age 216

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D1 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG37 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #217

1 Single Door #217

Open Space 206 to/from Of- 105° LH
fice 217

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D3 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG05 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #218

1 Single Door #218

Open Space 206 to/from Of- 105° LH
fice 218

Opening Description: 3' 0" x 8' 0" x 1 3/4" x WD Type WD-NA-D3 x HMF Type HM-NA-F1 x
N/A

4 Hinges	MPB79 4 1/2 x 4 1/2 NRP	10BE	MC
1 Lockset	70 10XG05 LL 2 3/4" BS	10BE	SA
1 Construction Core	7190224		BE
Note: - SFIC Construction core			
1 Wall Stop	409	10BE	RO
1 SFIC	33700006 N 26 DAKM		MED1
Note: - Product shall be US10B (or 10BE if available) finish per catalog selections			
- End user needs to confirm master key system & keyway prior procurement			
3 Door Silencer	608-RKW	BLK	RO

Set #MISC

1	Other Door #MISC	NH
---	------------------	----

1	Key Cabinet	KEYCAB	VA01
---	-------------	--------	------

Note: - Capacity of key cabinet shall be 150% of permanent cores supplied

2	Control Key	7147449	BE
---	-------------	---------	----

Note: - Construction core control key

10	Operating Key	7148872	BE
----	---------------	---------	----

Note: - Construction core operating key